Study of velocity increments in turbulent flows

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Abstract

We derive asymptotic approximations for velocity increment PDFs in the multifractal propagator framework. Using Laplace's method, we obtain two solutions: a simplified form assuming a fixed saddle point, and a better approximation via the Lambert W function. Numerical tests against the full propagator integral reveal where each approximation works best, providing groundwork for entropy calculations.

1 Introduction

Turbulent flows exhibit intermittency: velocity fluctuations become increasingly non-Gaussian at smaller scales. The fluctuations at scale ℓ can be studied with the velocity increments equation.

The multifractal approach describes intermittency through local Hölder exponents h distributed according to a spectrum $D_E(h)$. Chevillard et al. [1] proposed a propagator formalism where increment PDFs appear as integrals over the Hölder distribution, allowing for Gaussian large-scale statistics with small-scale intermittency.

We use the Kolmogorov–Oboukhov lognormal model with $D_E(h) = 1 - (h - c_1)^2/(2c_2)$. This gives tractable expressions while capturing intermittency. Since the propagator integrals lack closed forms, we apply Laplace's method for asymptotic approximations at small scales.

The paper proceeds as follows. Section 2 establishes the Hölder distribution and propagator integral. Section 3 derives two approximations of the increment PDFs: assuming a fixed saddle point (Case A), and then using the Lambert W function (Case B). We compare both approximations numerically against the full integral to assess their accuracy. Section 4 introduces standardized PDFs using the same pathway as in section 3.

2 Starting point: the distribution of h

The multifractal formalism for velocity increments can be interpreted as a continuous mixture of Gaussian variables. More precisely, the velocity increment at scale ℓ, denoted as $\delta_\ell u$, is modeled as a Gaussian random variable conditioned on a fluctuating singularity exponent h (Chevillard et al., 2012, Eq. 10) :

$$\delta_\ell u \overset{\text{law}}{=} \sigma \left(\frac{\ell}{L}\right)^h \delta \tag{0}$$

where σ represents the characteristic velocity of the flow at the integral scale L, δ is a zero-mean, unit-variance Gaussian white noise, and h is a fluctuating variable independent of δ, whose probability density function $\mathcal{P}_h^{(\ell)}$ is governed by a singularity spectrum $\mathcal{D}(h)$.

We now use (Chevillard et al., 2012, Eq. 11), which provides the probability density function of h :

$$\mathcal{P}_h^{(\ell)}(h) = \frac{\left(\frac{\ell}{L}\right)^{1-\mathcal{D}^E(h)}}{\displaystyle\int_{h_{\min}}^{h_{\max}} \left(\frac{\ell}{L}\right)^{1-\mathcal{D}^E(h)} \,\mathrm{d}h}. \tag{1}$$

using the Kolmogorov–Oboukhov log-normal spectrum

$$\mathcal{D}^E(h) = 1 - \frac{(h-c_1)^2}{2c_2},$$

with $h_{\min} = -\infty$ and $h_{\max} = +\infty$, we obtain

$$\mathcal{P}_h^{(\ell)}(h) = \frac{\exp\left[-\frac{(h-c_1)^2}{2c_2}\ln\left(\frac{L}{\ell}\right)\right]}{\displaystyle\int_{-\infty}^{+\infty} \exp\left[-\frac{(h-c_1)^2}{2c_2}\ln\left(\frac{L}{\ell}\right)\right] \mathrm{d}h}. \tag{2}$$

To simplify notation, we define

$$\alpha = -\ln\left(\frac{\ell}{L}\right) > 0. \tag{3}$$

Then the expression becomes

$$\mathcal{P}_h^{(\ell)}(h) = \frac{\exp\left[-\frac{\alpha}{2c_2}(h-c_1)^2\right]}{\displaystyle\int_{-\infty}^{+\infty} \exp\left[-\frac{\alpha}{2c_2}(h-c_1)^2\right] \mathrm{d}h}. \tag{4}$$

The denominator is a Gaussian integral:

$$\int_{-\infty}^{+\infty} e^{-\frac{\alpha}{2c_2}(h-c_1)^2} \mathrm{d}h = \sqrt{\frac{2\pi c_2}{\alpha}}. \tag{5}$$

Hence,

$$\boxed{\mathcal{P}_h^{(\ell)}(h) = \sqrt{\frac{\alpha}{2\pi c_2}}\, \exp\left[-\frac{\alpha}{2c_2}(h-c_1)^2\right].} \tag{6}$$

3 Non-normalized PDF

3.1 General form of the PDF

We now consider the probability density function of velocity increments. It's a function of the position x. According to (Chevillard et al., 2012, Eq. 17) :

$$\mathcal{P}_{\delta_\ell u}(x) = \frac{1}{2\pi\sigma}\sqrt{\frac{\alpha}{c_2}} \int_{-\infty}^{+\infty} \exp[-f(h,\alpha,x)]\,\mathrm{d}h. \tag{7}$$

with

$$f(h,\alpha,x) = \alpha\left(\frac{(h-c_1)^2}{2c_2} - h\right) + \frac{1}{2}\left(\frac{x}{\sigma}\right)^2 e^{2\alpha h} \tag{8}$$

We would like to get an explicit form of our Eq. 7. To do so, we'll use Laplace's approximation (see Appendix). Knowing that

$$\forall h \in \mathbf{R}, \quad \frac{\partial f}{\partial h}(h,\alpha,x) = \alpha\left(\frac{h-c_1}{c_2} - 1\right) + \alpha\left(\frac{x}{\sigma}\right)^2 e^{2\alpha h} \tag{9}$$

we have :

$$\forall h \in \mathbf{R}, \quad \frac{\partial^2 f}{\partial h^2}(h, \alpha, x) = \frac{\alpha}{c_2} + 2\alpha^2 \left(\frac{x}{\sigma}\right)^2 e^{2\alpha h} \quad > 0 \tag{10}$$

We just need to verify that $h^\star$ is such that $\frac{\partial f}{\partial h}(h^\star, \alpha, x) = 0$ to be able to use Laplace approximation.

In this case, by applying the approximation to Eq. 7, we get :

$$\boxed{\mathcal{P}_{\delta_\ell u}(x) \approx \sqrt{\frac{\alpha}{2\pi c_2 \sigma^2 \frac{\partial^2 f}{\partial h^2}(h, \alpha, x))}} \exp[-f(h^\star, \alpha, x)].} \tag{11}$$

i.e

$$\boxed{\mathcal{P}_{\delta_\ell u}(x) \approx \frac{1}{\sqrt{2\pi}\sqrt{2\alpha c_2 x^2 e^{2\alpha h^\star} + \sigma^2}} \times \exp\left[-\alpha\left(\frac{(h^\star - c_1)^2}{2c_2} - h^\star\right) - \frac{1}{2}\left(\frac{x}{\sigma}\right)^2 e^{2\alpha h^\star}\right]} \tag{12}$$

We provide plots of the obtained PDF using Julia programming language.

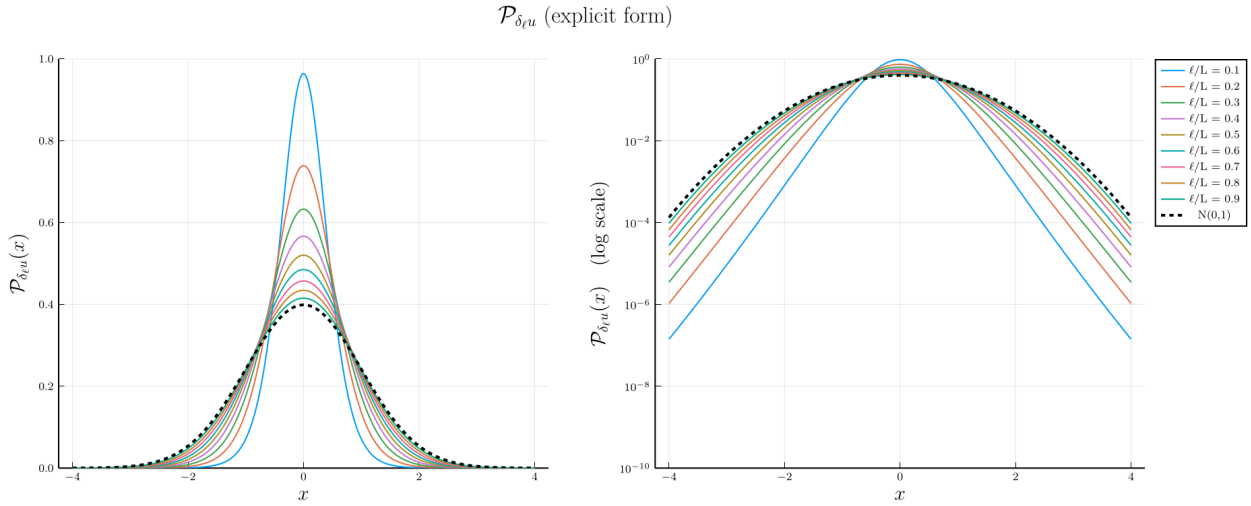


Figure 1: $\mathcal{P}_{\delta_\ell u}$ plots in linear and log scales using the explicit form of the PDF (Julia, QuadGK)

3.2 Case A: assuming $h^\star = c_1$

As a first approximation, we can consider that :

$$\frac{\partial f}{\partial h}(h^\star, \alpha, x)\Big|_{h^\star = c_1} = 0 \tag{13}$$

We can use the Laplace approximation given by Eq. 12, which becomes :

$$\boxed{\mathcal{P}_{\delta_\ell u}(x) \approx \frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{2\alpha c_2 x^2 e^{2\alpha c_1} + \sigma^2}} \times \exp\left[\alpha c_1 - \frac{1}{2}\left(\frac{x}{\sigma}\right)^2 e^{2\alpha c_1}\right]} \tag{14}$$

Here are plots of the obtained PDF using Julia programming language.

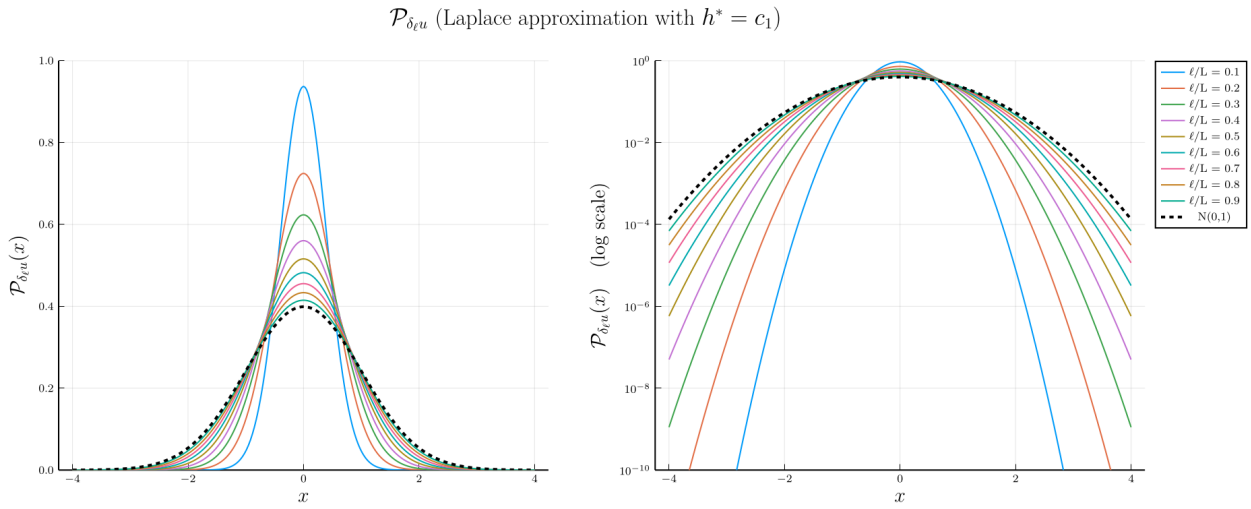


Figure 2: $P_{\delta_\ell u}$ plots in linear and log scales using Laplace approximation of the PDF with $h^\star = c_1$ (Julia, QuadGK)

3.3 Case B: solving $\frac{\partial f}{\partial h}(h, \alpha, x) = 0$ via the Lambert W function

From (8),

$$\frac{\partial f}{\partial h}(h, \alpha, x) = \alpha\left(\frac{h - c_1}{c_2} - 1\right) + \alpha\left(\frac{x}{\sigma}\right)^2 e^{2\alpha h} \tag{15}$$

Set to zero:

$$h - (c_1 + c_2) = -c_2 \left(\frac{x}{\sigma}\right)^2 e^{2\alpha h} \tag{16}$$

Multiply by -2α and let $u = -2\alpha(h - c_1 - c_2)$:

$$u = 2\alpha c_2 \left(\frac{x}{\sigma}\right)^2 e^{2\alpha(c_1+c_2)-u} \tag{17}$$

Consequently

$$ue^u = 2\alpha c_2 \left(\frac{x}{\sigma}\right)^2 e^{2\alpha(c_1+c_2)} \tag{18}$$

Hence

$$\boxed{h^\star(\alpha, x) = (c_1 + c_2) - \frac{1}{2\alpha} W\left(2\alpha c_2 \left(\frac{x}{\sigma}\right)^2 e^{2\alpha(c_1+c_2)}\right)} \tag{19}$$

where W is the Lambert W function (the inverse of $w \mapsto we^w$).

We can now plug the obtained $h^\star$ into Eq. 12. Here are plots of the obtained PDF using Julia programming language.

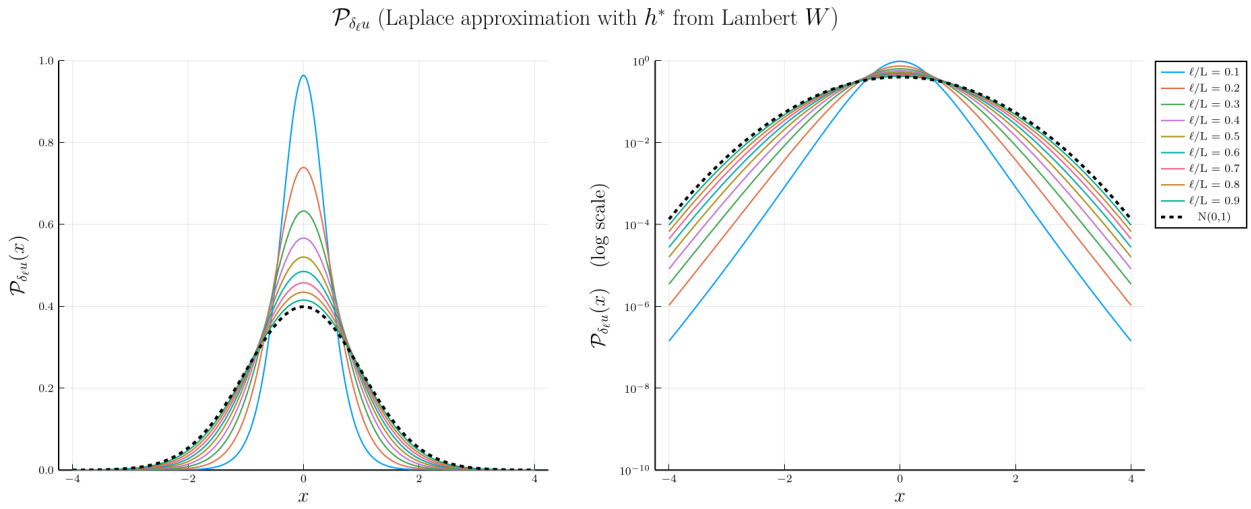


Figure 3: $\mathcal{P}_{\delta_\ell u}$ plots in linear and log scales using Laplace approximation with $h^\star$ from Lambert W (Julia, QuadGK)

Comparing the plots, it is clear that the approximation using the Lambert W function is better than the one assuming $h^\star = c_1$.

3.4 Properties of the non-normalized PDFs

We provide plots of the second-order structure functions and of the flatness functions in the three different cases.

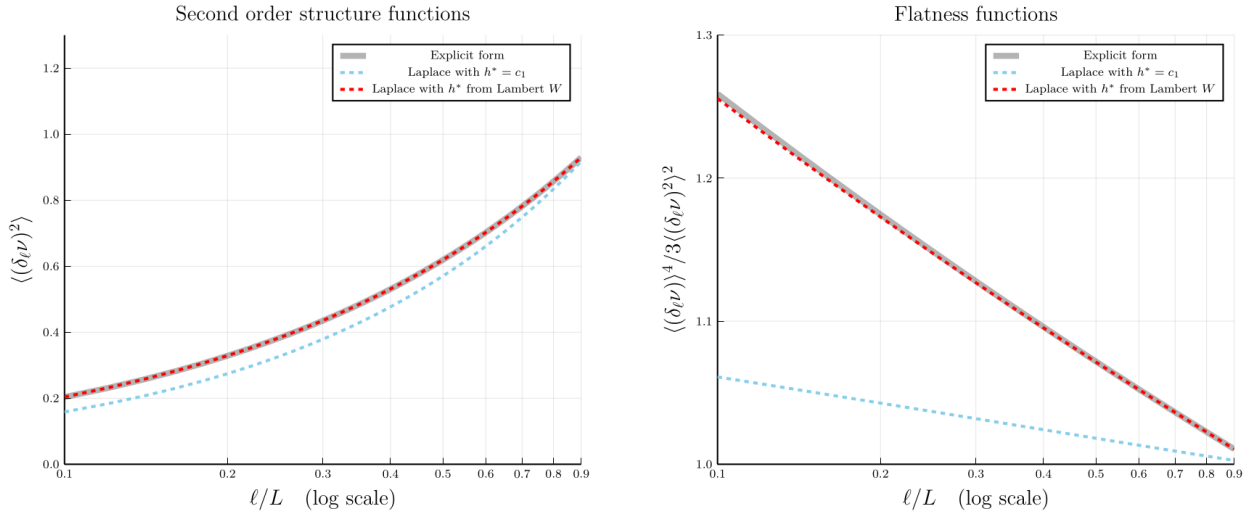


Figure 4: Second-order structure functions and flatness of $\mathcal{P}_{\delta_\ell u}$ in linear and log scales (Julia, QuadGK)

We also consider the Shannon entropy :

$$S(\delta_\ell u) = -\int_{-\infty}^{+\infty} \mathcal{P}_{\delta_\ell u}(x)\, \ln\big[\mathcal{P}_{\delta_\ell u}(x)\big]\, \mathrm{d}x \tag{20}$$

Here are plots of the Shannon entropy using the three different expressions of $\mathcal{P}_{\delta_\ell u}$.

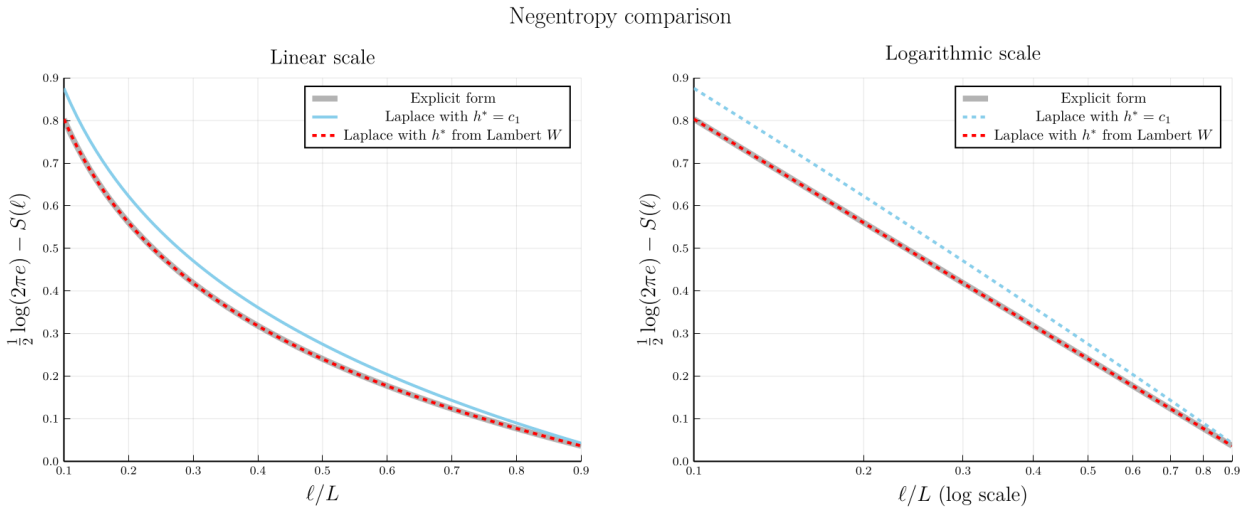


Figure 5: Shannon entropy of $\mathcal{P}_{\delta_\ell u}$ in linear and log scales (Julia, QuadGK)

4 Normalized PDF

According to (Chevillard et al., 2012, Eq. 12), the p-th order moment of the velocity increment is expressed as

$$\langle|\delta_\ell u|^p\rangle \underset{\ell\to 0}{\approx} \sigma^2\langle|\delta|^p\rangle \left(\frac{\ell}{L}\right)^{\zeta_p}, \quad \text{with} \quad \zeta_p = \min_h\big[ph + 1 - \mathcal{D}^E(h)\big] \quad \text{and} \quad \delta \sim \mathcal{N}(0,1). \tag{21}$$

Using the Kolmogorov–Oboukhov log-normal spectrum

$$\mathcal{D}^E(h) = 1 - \frac{(h-c_1)^2}{2c_2}, \tag{22}$$

the minimization problem is

$$\frac{\mathrm{d}}{\mathrm{d}h}\big[ph + 1 - \mathcal{D}^E(h)\big] = 0 \quad i.e. \quad \frac{\mathrm{d}}{\mathrm{d}h}\big[ph + \frac{(h-c_1)^2}{2c_2}\big] = 0 \tag{23}$$

Hence, the minimizing exponent is

$$h_p = c_1 - pc_2, \tag{24}$$

and inserting this value into Eq. 20 gives

$$\zeta_p = pc_1 - \frac{p^2}{2}c_2 \tag{24}$$

In particular, for the second-order :

$$\zeta_2 = 2(c_1 - c_2) \tag{26}$$

Knowing that $\langle|\delta|^2\rangle = 1$, we get :

$$\boxed{\langle(\delta_\ell u)^2\rangle \underset{\ell\to 0}{\approx} \sigma^2 \left(\frac{\ell}{L}\right)^{2(c_1-c_2)}}. \tag{27}$$

We now define a normalized increment

$$\boxed{\delta_\ell \nu = \frac{\delta_\ell u}{\sqrt{\langle(\delta_\ell u)^2\rangle}} = \frac{\delta_\ell u}{\sigma\left(\frac{\ell}{L}\right)^{c_1-c_2}}} \tag{28}$$

This normalization ensures that

$$\langle (\delta_\ell \nu)^2 \rangle = 1, \tag{29}$$

for all scales ℓ/L, so that all normalized PDFs can be directly compared across scales.

We thus have $\delta_\ell \nu \ = \delta_\ell u / \sigma \left(\frac{\ell}{L}\right)^{c_1 - c_2} = g(\delta_\ell u)$ where $g(x) = x/\sigma \left(\frac{\ell}{L}\right)^{c_1 - c_2}$ is monoton and differentiable.

Knowing that : $\mathcal{P}_{\delta_\ell \nu}(x) = \left| \frac{1}{g'(g^{-1}(x))} \right| \mathcal{P}_{\delta_\ell u}(g^{-1}(x))$, we have : $\mathcal{P}_{\delta_\ell \nu}(x) = \sigma \left(\frac{\ell}{L}\right)^{c_1 - c_2} \mathcal{P}_{\delta_\ell u}(\sigma \left(\frac{\ell}{L}\right)^{c_1 - c_2} x)$ i.e.

$$\boxed{\mathcal{P}_{\delta_\ell \nu}(x) = \sigma e^{\alpha(c_2 - c_1)} \mathcal{P}_{\delta_\ell u}(\sigma e^{\alpha(c_2 - c_1)} x)} \tag{30}$$

We then obtain from Eq. 12 the normalized form of the PDF :

$$\boxed{\mathcal{P}_{\delta_\ell \nu}(x) \approx \frac{e^{\alpha(c_2 - c_1)}}{\sqrt{2\pi}\sqrt{2\alpha c_2 x^2 e^{2\alpha(h^\star + c_2 - c_1)} + 1}} \times \exp\left[-\alpha \left(\frac{(h^\star - c_1)^2}{2c_2} - h^\star \right) - \frac{1}{2} x^2 e^{2\alpha(h^\star + c_2 - c_1)} \right]} \tag{31}$$

which is valid if $h^\star$ is such that $\frac{\partial f}{\partial h}(h^\star, \alpha, \sigma e^{\alpha(c_2 - c_1)} x) = 0$.

As in the non-normalized case, we provide the plots of the obtained PDF in linear ang log cases using Julia programming language.

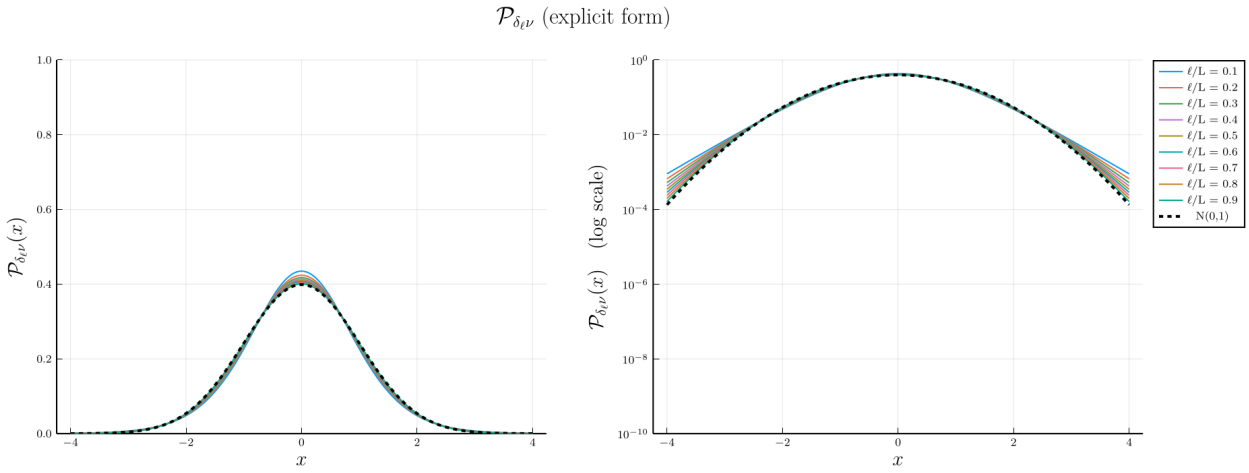


Figure 6: $P_{\delta_\ell \nu}$ plots in linear and log scales using the explicit form of the PDF (Julia, QuadGK)

4.1 Case A: assuming $h^\star = c_1$

If we consider that $\frac{\partial f}{\partial h}(h^\star, \alpha, \sigma e^{\alpha(c_2 - c_1)} x)\Big|_{h^\star = c_1} = 0$, then we get from Eq. 30 :

$$\boxed{\mathcal{P}_{\delta_\ell \nu}(x) \approx \frac{1}{\sqrt{2\pi}\sqrt{2\alpha c_2 \, x^2 e^{2\alpha c_2} + 1}} \times \exp\left[\alpha c_2 - \frac{1}{2} x^2 e^{2\alpha c_2} \right]} \tag{32}$$

Here are the plots the obtained PDF in linear and log scales using Julia programming language.

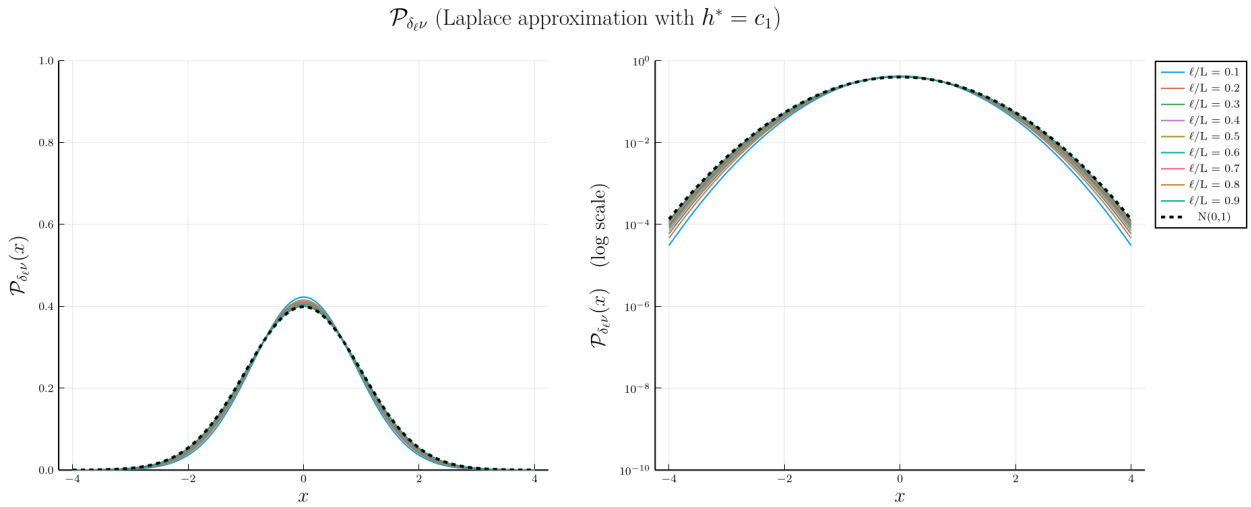


Figure 7: $P_{\delta_\ell\nu}$ plots in linear and log scales using Laplace approximation of the PDF with $h^* = c_1$ (Julia, QuadGK)

4.2 Case B: solving $\frac{\partial f}{\partial h}(h, \alpha, \sigma e^{\alpha(c_2-c_1)}x) = 0$ via the Lambert W function

We want to find $h^\star$ such that $\frac{\partial f}{\partial h}(h^\star, \alpha, \sigma e^{\alpha(c_2-c_1)}x) = 0$. Thanks to Eq. 19, we have :

$$\boxed{h^\star(\alpha, \sigma e^{\alpha(c_2-c_1)}x) = (c_1 + c_2) - \frac{1}{2\alpha}W\big(2\alpha c_2 x^2 e^{4\alpha c_2}\big)} \tag{33}$$

We can now plug the obtained $h^\star$ into Eq. 30. Let $W = W\big(2\alpha c_2 x^2 e^{4\alpha c_2}\big)$. We get :

$$\boxed{\mathcal{P}_{\delta_\ell\nu}(x) \approx \frac{1}{\sqrt{2\pi}\sqrt{2\alpha c_2 x^2 e^{4\alpha c_2 - W} + 1}} \times \exp\left[-\frac{\alpha}{2}\left(\frac{1}{c_2}\left(\frac{W}{2\alpha}\right)^2 - 3c_2\right) - \frac{1}{2}x^2 e^{4\alpha c_2 - W}\right]} \tag{34}$$

i.e.

$$\boxed{\mathcal{P}_{\delta_\ell\nu}(x) \approx \frac{e^{\frac{3}{2}}c_2\alpha}{\sqrt{2\pi}}\frac{e^{-\frac{2W+W^2}{8c_2\alpha}}}{\sqrt{1+W}}} \tag{35}$$

which is the same equation as (Granero Belinchón et al., 2017, Eq. 18).

We plot again the obtained PDF in linear and log cases using Julia programming language.

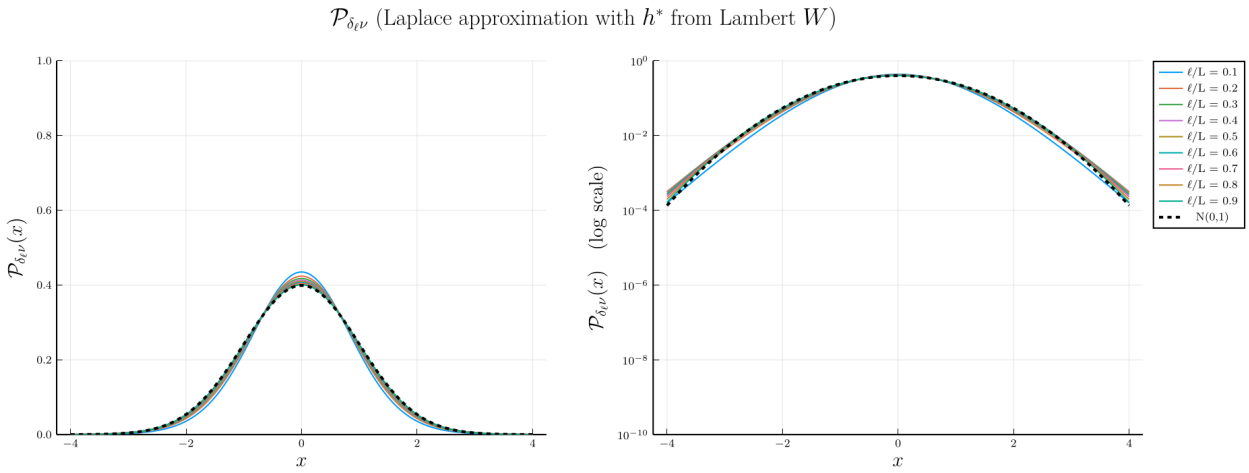


Figure 8: $P_{\delta_\ell\nu}$ plots in linear and log scales using Laplace approximation with h^* from Lambert W (Julia, QuadGK)

4.3 Properties of the normalized PDFs

As in the non-normalized case, we provide plots of the second-order structure functions and of the flatness functions in the three different cases.

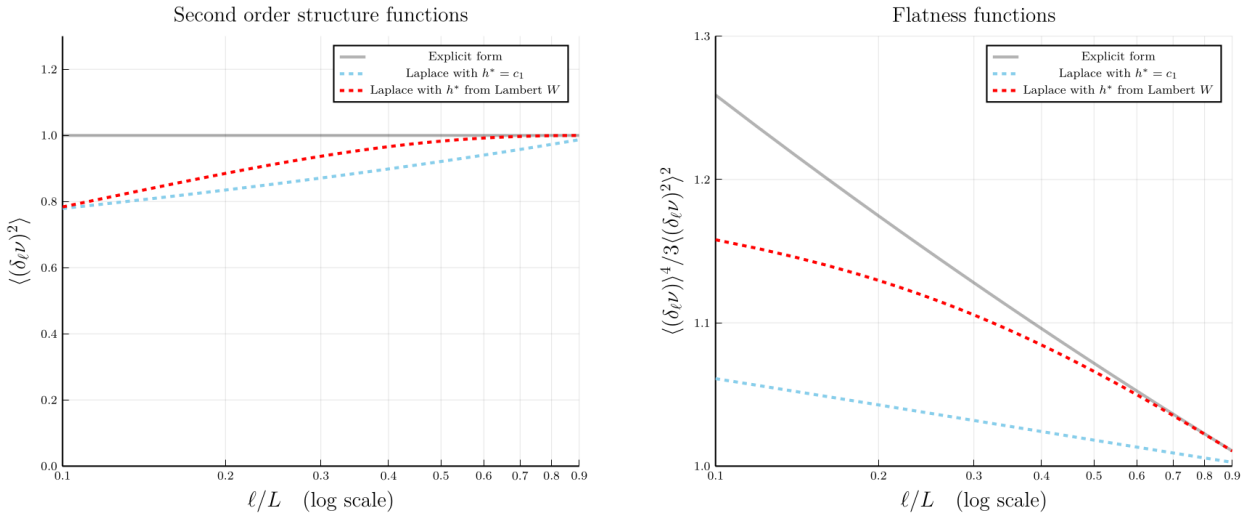


Figure 9: Second-order structure functions and flatness of $\mathcal{P}_{\delta_\ell\nu}$ in linear and log scales (Julia, QuadGK)

We consider the Shannon entropy in the normalized case :

$$S(\delta_\ell\nu) = -\int_{-\infty}^{+\infty} \mathcal{P}_{\delta_\ell\nu}(x)\, \ln\big[\mathcal{P}_{\delta_\ell\nu}(x)\big]\, \mathrm{d}x \tag{36}$$

Here are plots of the Shannon entropy using the three different expressions of $\mathcal{P}_{\delta_\ell\nu}$.

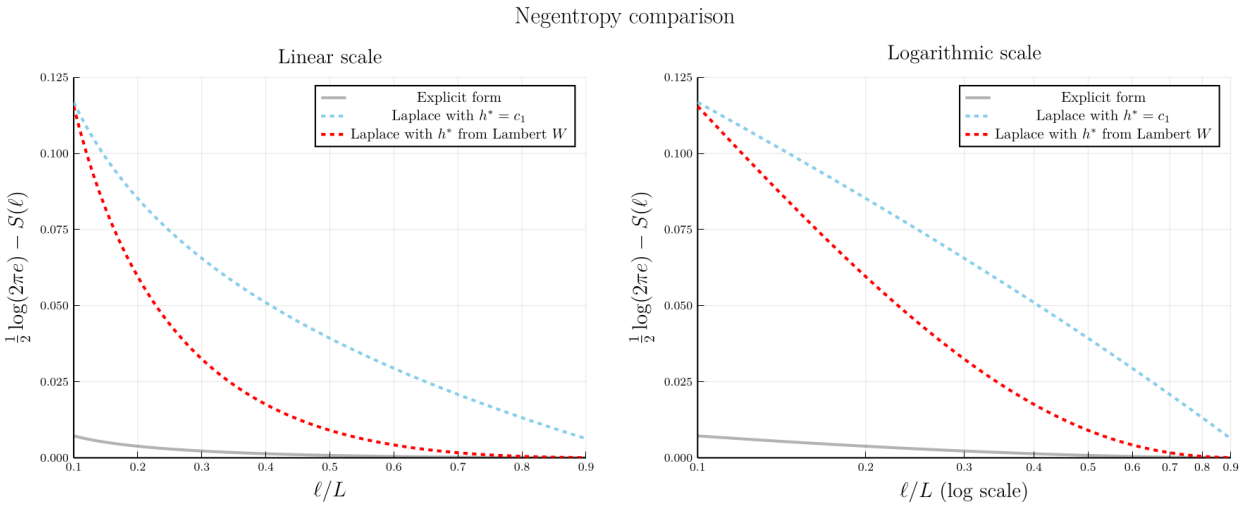


Figure 10: Shannon entropy of $\mathcal{P}_{\delta_{\ell\nu}}$ in linear and log scales (Julia, QuadGK)

References

[1] L. Chevillard, B. Castaing, A. Arneodo, E. Lévêque, J.-F. Pinton, and S. G. Roux, *A phenomenological theory of Eulerian and Lagrangian velocity fluctuations in turbulent flows*, C. R. Physique **13**, 899–928 (2012).

[2] C. Granero-Belinchón, S. G. Roux, and N. B. Garnier, *A Kullback-Leibler divergence measure of intermittency: application to turbulence*, Phys. Rev. E **97**, 013112 (2017).

A Appendix - Laplace's approximation theorem

Let $I = [a, b]$ be an interval, $\varphi \in C^2(I, \mathbb{R})$ and $g \in C^0(I, \mathbb{C})$. Assume that:

(i) For all $t > 0$, $\displaystyle\int_a^b e^{t\varphi(x)}|g(x)|\,dx < +\infty$;

(ii) φ' vanishes at a unique point $x_0 \in I$ and $\varphi''(x_0) < 0$;

(iii) $g(x_0) \neq 0$.

Then

$$\int_a^b e^{\,t\varphi(x)}\,g(x)\,dx \sim \sqrt{\frac{2\pi}{t\,|\varphi''(x_0)|}}\,e^{\,t\varphi(x_0)}\,g(x_0) \qquad \text{as } t \to +\infty.$$